

[Olive pollinosis in Japan].

PMID: 8857110 | Indexed in PubMed

Olive (*Olea europaea*) is the most important allergic tree in the Mediterranean area. That is widely distributed at Syodosima in Japan. I observed olive pollination period and studied the patients with pollinosis. The patients were examined by skin test, measuring of specific IgE antibody and nasal provocation test with olive pollen extract. Allergic responses to olive pollen were found to be present in 16.3% of pollinosis patients. And cross-reactivity between olive and grass pollen (*Dactylis glomerata*) was suggested. Other literature reported the cross reactivity between the pollens of the Oleaceae family, containing olive, and grass family. From our study we suppose that Oleaceae pollens cause allergic reaction of Gramineae pollinosis patients in Japan.

Serum IgE profiles in Chinese pollinosis patients with grass pollen sensitisation.

PMID: 35079319 | Indexed in PubMed

Pollen from trees, grasses, and weeds is a common allergen source. The characteristics of pollen allergy in China are obviously different from Europe. Most studies have focused on tree and weed pollen, but there is a paucity of data on grass pollen sensitisation in China. Therefore, we used component-resolved diagnostics to investigate the serum-specific immunoglobulin E (sIgE) to grass pollen in Chinese patients with pollinosis. We retrospectively analysed 547 patients with pollen allergy from an outpatient Allergy Department in Beijing, China. All the patients answered questionnaires about their clinical allergy histories. Total immunoglobulin E (IgE) and sIgE levels to grass pollen (Bermuda, Timothy grass) were quantified by ImmunoCAP using 0.35 kUA/L as a threshold for positivity. Of the 547 pollinosis patients, 389 (71.1%) showed a positive sIgE reaction to either grass pollen, or both. The prevalence of food allergy was significantly lower in patients with grass pollen sensitisation. Among the 389 patients with grass pollen sensitisation, the prevalence of sIgE to allergen extracts of bermuda, mugwort, ragweed, plane, hop, ash, birch, and timothy grass was 97%, 96%, 94%, 88%, 88%, 84%, 78%, and 78%, respectively. However, only 134/389 (34%) were positive for Cyn d 1, 29/389 (7%) for Phl p 1, and 8/389 (2%) for Phl p 5b. For pollinosis patients, 62/547 (11%) were sIgE-positive for cross-reactive carbohydrate determinants (CCDs), and their grass pollen-sIgE was also positive. The prevalence of in vitro IgE sensitisation to grass pollen extract is high in Chinese patients with pollinosis. But mostly spurious and characterized by IgE sensitisation to profilins and CCD, induced by other pollen. Component-resolved diagnostics is an extremely useful tool precise diagnostics of pollen allergy in China.

Pollen-related allergy in Italy.

PMID: 1586007 | Indexed in PubMed

Pollen-related allergies are very common in Italy and pollinosis is the commonest allergic disease. The type of allergenic plants and the prevalence of hay fever varies among regions. In the Mediterranean area there are characteristic climatic conditions (mildness of winter, summer dryness) that facilitate the growth of a typical vegetation with its associated various types of allergenic pollen grains, some of them very different from those of central and northern Europe. Italy has a central position in the Mediterranean basin, but because of its geographic characteristics, there are different climatic aspects with different

vegetation between northern, central, and southern areas. Gramineae are the most common allergenic plants in northern and central Italy, where more than 60% of patients with pollinosis are grass-pollen sensitive. Parietaria is the most important pollinating plant in southern Italy and Liguria. Olea europaea, the olive tree with cultivation widespread in the whole Mediterranean basin, is responsible for frequently severe pollinosis, particularly in some regions of the southern Italy.

Antigen-specific immunotherapy against allergic rhinitis: the state of the art.

PMID: 20093851 | Indexed in PubMed

Allergic rhinitis is the most prevalent type I allergy in industrialized countries. Pollen scattering from trees or grasses often induces seasonal allergic rhinitis, which is known as pollinosis or hay fever. The causative pollen differs across different areas and times of the year. Impaired performance due to pollinosis and/or medication used for treating pollinosis is considered to be an important reason for the loss of concentration and productivity in the workplace. Antigen-specific immunotherapy is an only available curative treatment against allergic rhinitis. Subcutaneous injection of allergens with or without adjuvant has been commonly used as an immunotherapy; however, recently, sublingual administration has come to be considered a safer and convenient alternative administration route of allergens. In this review, we focus on the safety and protocol of subcutaneous and sublingual immunotherapy against seasonal allergic rhinitis. We also describe an approach to selecting allergens for the vaccine so as to avoid secondary sensitization and adverse events. The biomarkers and therapeutic mechanisms for immunotherapy are not fully understood. We discuss the therapeutic biomarkers that are correlated with the improvement of clinical symptoms brought about by immunotherapy as well as the involvement of Tr1 and regulatory T cells in the therapeutic mechanisms. Finally, we focus on the current immunotherapeutic approach to treating Japanese cedar pollinosis, the most prevalent pollinosis in Japan, including sublingual immunotherapy with standardized extract, a transgenic rice-based edible vaccine, and an immunoregulatory liposome encapsulating recombinant fusion protein.

Pollen counts in Slavonski Brod, Croatia during the pollination period 2008 to 2010.

PMID: 22450204 | Indexed in PubMed

Hay fever or pollinosis is the allergic reaction of the human body to allergic pollen grains and is a seasonal phenomenon. Pollen concentrations depend on the climate, geographic features, and vegetation. Trees, grass, and weed pollinosis is frequent in Croatia, common ragweed (*Ambrosia artemisiifolia* L.) pollinosis in particular. Continuous monitoring of pollen air concentrations can provide timely information to the general public and can help sensitised patients and their physicians to prevent or alleviate allergic reactions. This is the task of health ecology services such as our Public Health Institute of the Brod-Posavina County. This article reports pollen concentrations in Slavonski Brod measured in March 2008 to November 2008, March 2009 to October 2009, and April 2010 to October 2010 and discusses the increasing exposure to ragweed and ways to control it.

[Pollen allergy and immunotherapy].

PMID: 22477663 | Indexed in PubMed

Pollinosis affects up to 25 % of the Swiss-population and increased substantially during the last century. Main causative pollens are birch and related pollens, ash pollen and grass pollen and to a lesser extent also mugwort. Diagnosis of Pollinosis is based on anamnesis, skin tests and determination of specific IgE in the serum. Treatment includes allergen avoidance, pharmacotherapy and allergen-specific immunotherapy (SIT). A proper patient and allergen extract selection, an ideally adapted application regimen and good patient-guidance as well as good compliance and treatment adherence are decisive for the success of the SIT. SCIT is the regimen of choice with well demonstrated efficacy, safety and long term effect. Nowadays SLIT is an additional complementary approach that mainly for grass pollen also offers good efficacy with good safety. Under such conditions immunotherapy leads to a benefit in more than 80 % of pollinosis-patients. Component-resolved diagnosis is a very useful and important step in more specific diagnosis. Several approaches studying changed application regimens and using molecular technologies to improve the efficacy of SIT are undergoing and will possibly substantially improve the efficacy, safety and acceptance of SIT.

[Pollinosis: clinical aspects and epidemiology. Contribution of the Allergy Clinic 1948-1998].

PMID: 9816923 | Indexed in PubMed

Pollinosis or hay fever is the most common allergic disease in Switzerland and also in the patients of the Allergy Unit of the Dermatologic Department of Dermatology of the University Hospital of Zurich. Clinical and epidemiological research concerning pollinosis has therefore always taken an important place, especially under Brunello Wüthrich. The most important clinical symptoms are seasonal conjunctivitis, rhinitis and in about 25% in a later stage also asthma. Pollinosis in central part of Switzerland is mainly caused by pollens of birch and related trees (alder, hazel), by pollen of ash and by pollen of grasses, rye and mugwort. The amount of measurable pollen is highly depending on geographic and climatic conditions and varies therefore considerably between different regions in Switzerland as well as different nations and continents. Hay fever has very much increased in the last decades; in the SAPALDIA study (Swiss Study on Air Pollution and Lung Diseases in Adults) a prevalence of up to 14% has been found recently. There are many causes not yet fully understood: genetics, different pollen exposure and some patterns of air pollution are discussed. A modern treatment of hay fever bases on prophylactic measures, symptomatic therapy with the now available efficient drugs with minimal side effects (topical drugs, oral antihistamines) and the specific immunotherapy.

Prevalence of pollinosis in patients with allergic asthma, rhinitis and conjunctivitis in the South of Mexico City 2007-2013.

PMID: 25177850 | Indexed in PubMed

The prevalence of pollinosis has doubled in the past two decades. Several studies suggest that up to 50% of adult residents of Mexico City can present manifestations of respiratory allergy, and pollens from trees, grasses and weeds are a common cause. To determine the prevalence of their families and antigenic cross-reactivity allows us to offer appropriate diagnoses and treatments. To know the prevalence of sensitization of pollens to trees, grasses and weeds in adults with respiratory allergy of the South zone of

Mexico City from January 2007 to December 2013. A cross-sectional, observational and prospective study was done with patients from Mexico City, referred to the National Medical Center Siglo XXI, IMSS, from 2007 to 2013 with a diagnosis of allergic rhinitis, asthma and conjunctivitis. We analyzed the results of skin prick tests to pollens from trees, grasses and weeds in selected patients. The results were analyzed using descriptive statistics. A total of 672 patients were analyzed, 70% men, the average age was 34 ± 16 years. Regarding occupation 31% were students, 48% employees and 21% housewives. Fifty-three percent had rhinitis, 47% had asthma and 40.5% had both, asthma and rhinitis. Prevalence of sensitization to weeds was 56%, 33% to trees and 11% to grasses. Sensitization to weeds is the first cause of respiratory pollinosis in the south of Mexico City, *Amaranthus* was the most prevalent pollen in this area. Sensitization to trees is the second cause, with a predominance of trees from Betulaceae, Fagaceae and Oleaceae families. Sensitization to grass is the third cause of respiratory pollinosis. The most common are from Poaceae (*Lolium perenne*), Chloridoideae and Cynodon/Dactyloctenium family.

Molecular and immunochemical characterization of Pop n 2: A new allergen of *Populus nigra* pollen.

PMID: 33210791 | Indexed in PubMed

Pollen is one of the most common allergens that cause respiratory allergies worldwide. Pollen grains from poplars have been reported as important sources of pollinosis in many countries. The aim of the present study was to determine the molecular and immunochemical characterization of Pop n 2, a novel allergen of *Populus nigra* (*P. nigra*) pollen extract. In this study, the pollen extract of *P. nigra* was analysed by SDS-PAGE, and the allergenic profile was determined by IgE immunoblotting and specific ELISA using the sera of twenty allergic patients. The coding sequence of Pop n 2 was cloned and expressed in the *Escherichia coli* BL21 (DE3) using plasmid the pET-21b (+). Finally, the expressed recombinant Pop n 2 was purified by affinity chromatography. Pop n 2 belongs to the profilin family with a molecular weight of approximately 14 kDa. Pop n 2 is the most IgE-reactive protein (about 65%) in the *P. nigra* pollen extract. The cDNA sequencing results indicated an open reading frame 396 bp that encodes 131 amino acid residues. The results of ELISA and Immunoblotting assays showed that recombinant Pop n 2 could react with the IgE antibody in patients' sera, like its natural counterpart. Our data revealed that Pop n 2 is a significant allergen in the *P. nigra* pollen extract. Moreover, we observed that the recombinant Pop n 2 produced by the pET-21b (+) vector in the *E. coli* system acts as its natural counterpart.

Allergy to Rosaceae fruits without related pollinosis.

PMID: 9438478 | Indexed in PubMed

Rosaceae fruit allergy is frequently associated with birch pollinosis in Central and Northern Europe and with grass pollen allergy in Central Spain. The main cross-reactive structures involved for birch pollinosis are Bet v 1 and profilin, and for grass pollinosis they are profilin and carbohydrate determinants. Rosaceae fruit allergy can occasionally be observed in patients without pollinosis. We investigated the clinical presentation and the allergens involved in allergy to Rosaceae fruit without pollinosis. Eleven patients from Central Spain allergic to apples, peaches, and/or pears but not to pollens were compared with 22 control subjects with combined grass pollen and fruit allergy. Skin prick tests and RASTs to apple, peach, and pear were performed. Cross-allergenicity was studied by RAST inhibition. Bet v 1 was tested

with an indirect RAST, and profilin was tested in skin prick tests, histamine release, and RAST. Rosaceae fruit allergy without pollinosis is severe with 82% of patients reporting systemic symptoms, mainly anaphylaxis (73%), whereas oral symptoms are less frequent (64%). Anaphylactic shock was observed in 36% of patients. The fruit allergens involved showed cross-reactivity among Rosaceae species but were not related to profilin or Bet v 1. Ninety-one percent of patients with combined grass pollinosis and fruit allergy reported oral allergy, 45% reported systemic symptoms, 18% reported anaphylaxis, and 9% reported anaphylactic shock. Allergy to Rosaceae fruits in patients without a related pollen allergy is a severe clinical entity. Profilin- and Bet v 1-related structures are not involved in Rosaceae fruit allergy without pollinosis.
